

Technical Specifications: OpenSourceSDRLab 7020-AD9361 (Advanced PlutoSDR)

Radiofrequency Engineering

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1 Gains

Managed through the `libiio` API (Linux Industrial I/O), which directly controls the AD9361 radio chains and the additional hardware integrated on this board:

- **RX Gain (AD9361):** Range from 0 to 73 dB. Can operate in MGC (Manual Gain Control) or through AGC (Automatic) with Fast, Slow, or Hybrid profiles. Affects the integrated LNA, mixer, and TIA filters.
- **TX Attenuation (AD9361):** Digitally adjustable attenuation from 0 to -89.75 dB in fine 0.25 dB steps.
- **Integrated PA (External Hardware):** Unlike the official PlutoSDR, this board includes a power amplifier (PA) in the transmit chain.

Direct on/off control of the PA depends on the GPIO pin configuration enabled in the modified firmware. By default, it significantly increases output power compared with the native ≈ 0 dBm of the AD9361.

2 Hardware Filters and Processing

- **Analog Processing (AD9361):** Transimpedance-amplifier (TIA) and baseband low-pass filters (LPF), highly configurable in real time to prevent aliasing and limit the analog channel bandwidth (from 200 kHz to 56 MHz).
- **Integrated Digital Filtering (AD9361):** 128-tap FIR filter blocks and CIC filters for interpolation/decimation before sending samples to the SoC.
- **SoC and FPGA (Xilinx Zynq XC7Z020-CLG400):** This is the largest jump compared with the PlutoSDR (which uses the 7010). The 7020 offers ≈ 3 times more logic cells (85k vs 28k), allowing complex custom bitstreams to be loaded (e.g., complete modems in VHDL/Verilog) and allowing the board to emulate a ZEDBOARD + FMCOMMS2/3 environment.

3 Frequency Ranges

- **Operating Range:** 70 MHz to 6.0 GHz.
- **Instantaneous Bandwidth:** Up to 56 MHz.
- **Hardware Note:** The board uses an AD9361 transceiver (or an AD9363 officially software-unlocked to behave like a 9361), which removes the calibration bottlenecks of low-cost commercial versions that operate only between 325 MHz and 3.8 GHz.

4 Sample Rate and Resolution

Driven by the AD9361 internal converters and the upgrade to **1 GB of DDR3 RAM** (twice that of the original PlutoSDR).

Data-throughput note: The standard PlutoSDR saturates quickly over USB 2.0 (practical limit ≈ 15 MS/s). Using the **Gigabit Ethernet** interface included on this board allows much higher sample rates to be sustained in network mode, approaching the RF chip limit.

Architecture	ADC/DAC Resolution	Max. Rate	Sample	Host Interfaces
2T2R (dual MIMO)	12-bit	61.44 MS/s		USB 2.0 / Gigabit Ethernet

5 IQ & DC Corrections

- **Internal Calibration:** The AD9361 transceiver has internal state machines that perform automatic DC offset tracking and correction (Tx and Rx) and quadrature imbalance correction (QEC).
- **BIST Phase (Built-In Self Test):** Calibrations run during initialization and can be forced to rerun through the `libiio` API in case of drastic temperature changes.

6 Connectors and Inputs

Connector	Function	Specification
Gigabit Ethernet	Network and high-speed control	RJ45. Ideal for heavy continuous streaming. (Warning: some firmware images disable USB when Ethernet is used).
Micro-USB / Type-C	Power and OTG	Classic connection for gadget mode (RNDIS) or USB peripherals.
RF (SMA) $\times 4$	TX1, TX2, RX1, RX2	50 Ω . Enables simultaneous 2×2 MIMO topologies.
REF CLK (SMA)	Reference Clock	Input for injecting high-precision 10 MHz or 40 MHz (GPSDO).
LO IN (SMA)	External Local Oscillator	Phase input for strict coherence between multiple boards.
JTAG	FPGA debugging / UART	Explicitly exposed. Includes serial lines for the boot console.

7 Safe Operating Limits

Critical warnings - front end and hardware modifications

- **Maximum RX power:** Do not exceed the theoretical **+2.5 dBm** (recommended 0 dBm maximum) on the RX1/RX2 ports to protect the AD9361 LNAs.
- **Closed TX-RX loop (Loopback):** Since this board **includes a PA in transmission**, the TX port will send much more energy than a conventional Pluto. Use attenuators of at least 40–50 dB in direct tests.
- **Reference Voltages:** Ensure that signals injected into REF CLK and LO IN meet the tolerable voltage levels (typically 1.8 V logic or capacitively coupled).
- **Heat Dissipation:** The Zynq 7020 and AD9361 generate significant heat, especially under MIMO and high sample rates. Requires a heat sink and aluminum enclosure or forced ventilation.

8 Clock and Advanced Synchronization (MIMO)

- Unlike the original basic design, this OpenSourceSDRLab board introduces **explicit REF CLK and LO (Local Oscillator) inputs**.
- **Phase Coherence:** Connecting the same Local Oscillator (LO) to several 7020-AD9361 boards allows them to be synchronized to form larger-scale antenna arrays (4×4 , 8×8), which is fundamental in phased-array radar or beamforming research (for which MATLAB or GNU Radio is used).

9 Indicator LEDs

LED	Function	Normal state
PWR	Power	Steady after powering on the unit.
DONE	FPGA Boot Status	Turns on when the Zynq correctly loads the firmware bitstream.
TX/RX	Radio activity	Blinking/steady to indicate operation of the AD9361 ports.
ETH Link	Gigabit port	Standard RJ45 signaling (Link/Act).

10 Expansion Interfaces (JTAG and GPIO)

Multipurpose JTAG Port

The manufacturer explicitly exposed this port to facilitate “bare-metal” development (without Linux) or to accelerate HDL simulations:

- Allows code to be loaded onto the dual-core ARM Cortex-A9 or the FPGA without going through the operating system.
- Includes pins for the UART console, used for U-Boot diagnostics during boot and IP configuration.

GPIO Expansion

Headers have been introduced with multiple GPIO pins connected directly to the FPGA logic (XC7Z020 pin bank). Ideal for:

- Controlling additional external amplifiers.
- Synchronizing relays and transmit/receive switches (T/R switches).
- Hardware triggers to/from other laboratory equipment.